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Progress Report

Description of Data

The data I will be using is compiled of 1000 patients and five variables which are typically associated with PCOS diagnosis and its risks. These variables are age, BMI, menstrual irregularity, testosterone level, and number of antral follicles. Finally, the response variable is whether a patient has been diagnosed with PCOS or not, making it a classification data set.

Variable	Description
Age	The age of the patient in years
BMI	weight(kg)/ height(m) ^{2}
Menstrual irregularity	0 means the patient has regular menstruation and 1 means they have irregular menstruation. Irregular menstruation could be classified as missing periods, heavy bleeding, or painful periods.
Testosterone level	Measured as nanograms per deciliter (ng/dL)
Antral follicle count	Antral follicles are the follicles on the patient's ovaries that contain immature eggs. This is the count of how many the patient has at the time of diagnosis.
PCOS diagnosis	0 means they were not diagnosed with PCOS while 1 means they were.

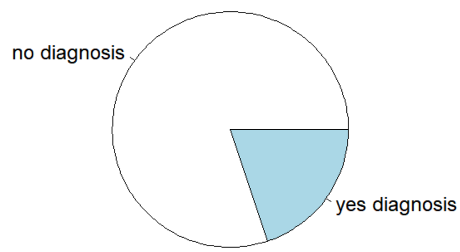
Processing Data

Range of variables:

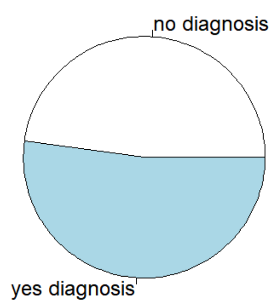
```
Age                         18.0 45.0
BMI                         18.2 35.0
Menstrual_Irregularity       0.0  1.0
Testosterone_Level.ng.dL.   20.1 99.6
Antral_Follicle_Count        5.0 29.0
PCOS_Diagnosis               0.0  1.0
```

Checking for NA's and balancing data:

After running a few tests, I see that I have no missing data throughout the 1000 entries and 6 variables. I do, however, have unbalanced data, 80.1% with no diagnosis and 19.9% with a diagnosis. Having such a large unbalance in data can lead to biased outcomes when trying to model the data.



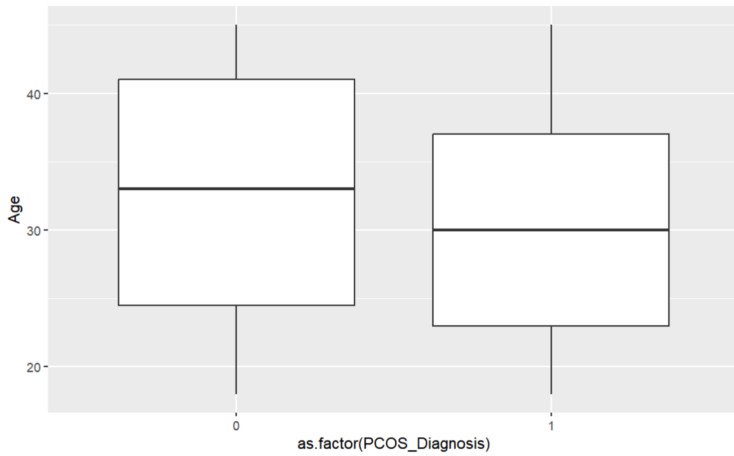
Now I have more balanced data with only 382 entries, 183 (47.9%) with no diagnosis and 199 (52.1%) with a diagnosis. Also, the range of the cut-down data is the same as the original. So, this data will be a great sample that does not lead to bias.



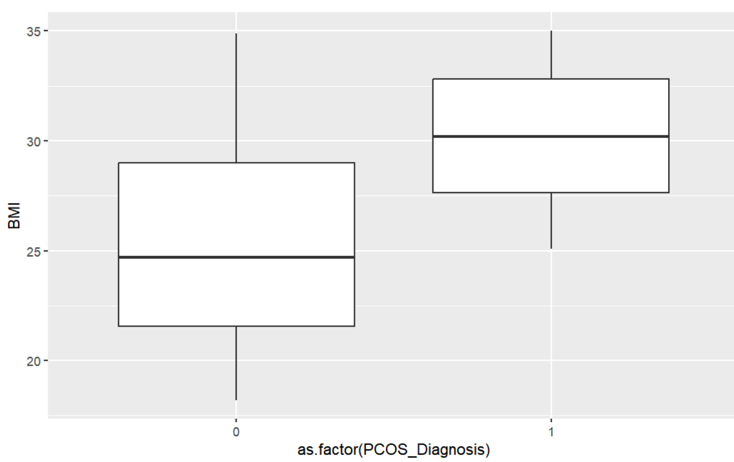
Age	18.0	45.0
BMI	18.2	35.0
Menstrual_Irregularity	0.0	1.0
Testosterone_Level.ng.dL.	20.1	99.6
Antral_Follicle_Count	5.0	29.0
PCOS_Diagnosis	0.0	1.0

Data Visualization

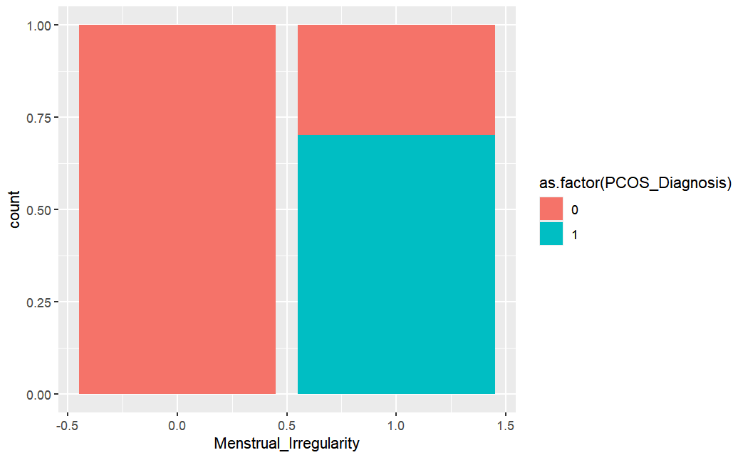
Now I will look at box plots of each explanatory variable against my response variable, PCOS diagnosis.



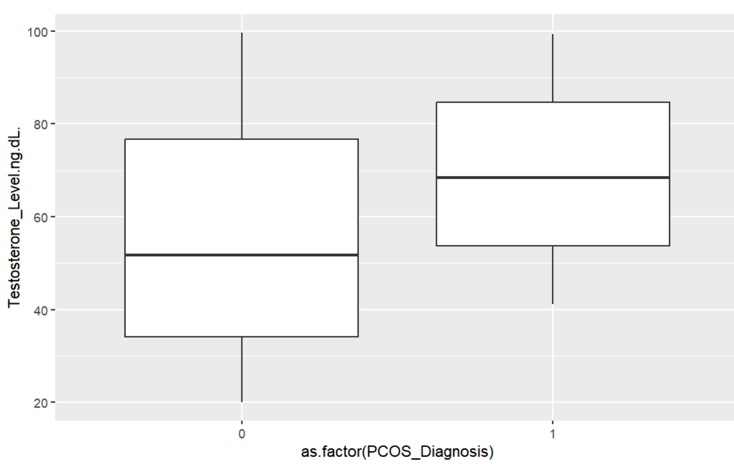
Here we can see that there is a slight difference in age and PCOS diagnosis. It seems women around 30 are more likely to get a PCOS diagnosis while women over 30 do not. But, still, this is only a very small difference and I may find out that age is not significant to PCOS diagnosis.



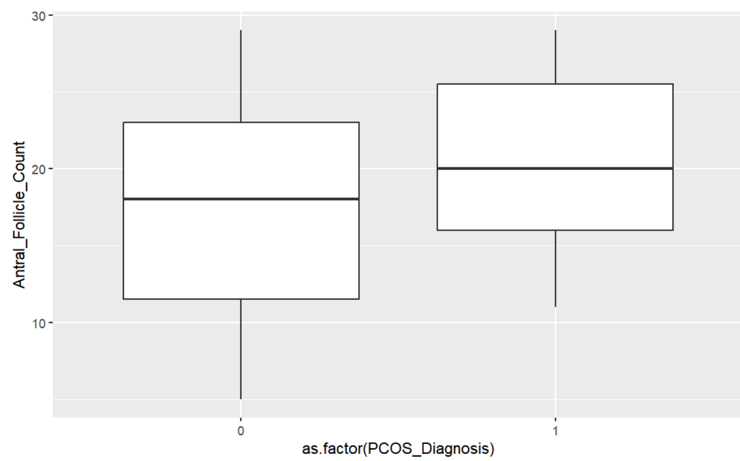
There is an obvious difference in BMI scores and PCOS diagnosis. It seems that the higher BMI score, the more likely you are to be diagnosed with PCOS.



Since menstrual irregularity is a categorical variable, I thought it best to show its effect against PCOS with a bar graph. Here we can see women with no menstrual irregularity (0) do not get diagnosed with PCOS. On the other hand, about 70% of the women who did report menstrual irregularity (1) were diagnosed with PCOS.



Same as BMI, those with higher testosterone levels were more likely to receive a PCOS diagnosis.



Lastly, there is a slight difference in the antral follicle count and PCOS diagnosis. Those with higher follicle count were more likely to receive a PCOS diagnosis. Although, perhaps I am to discover that this variable is not significant.

Choosing Variables for the Models

Logistic Regression:

Since the data is categorical (0 for no diagnosis, 1 for a diagnosis), it can be fitted to a logistic regression model to see which variables in my data are most significant to whether someone receives a PCOS diagnosis. After fitting the five variables to PCOS_Diagnosis, their p-values show that age and menstrual irregularity are considered to be statistically insignificant, while BMI, testosterone level, and antral follicle count are considered significant. This is a bit surprising since menstrual irregularity is considered to be a main symptom from PCOS.

Backward Selection:

Another method to find the best collection of variables is through backward selection. Backward selection works by having all the variables in the model and taking out one one-at-a-time until there is no further improvement in the AIC, meaning the AIC no longer shrinks.

After applying this method to the model with the original variables, it shows that the best combination of variables are all of the predictors except age, having an AIC of 178.49.

Forward Selection:

Forward selection begins with a null model and, similar to backwards selection, adds one variable one-at-a-time until the AIC lowers without any further improvement.

Using forward selection, we get the same combination of parameters with a 178.49 AIC.

The outcome of both of these tests makes sense since age is not a factor in whether or no someone has PCOS, but rather indicates the time when they went in for a diagnosis.

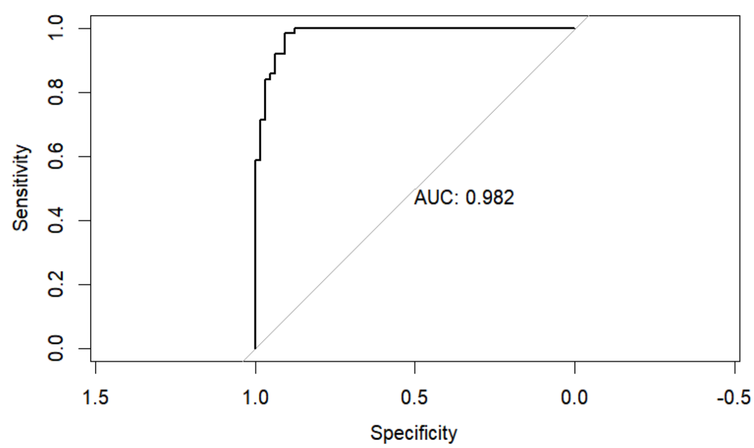
From these two methods, I have concluded that the best set of parameters to predict PCOS diagnosis are: Age, BMI, Menstrual_Irregularity, Testosterone_Level, and Antral_Follicle_Count.

Testing with Different Models

Before testing can begin, the data must be split into training and testing data for comparison. The training data consisted of a random sample of ⅔'s of the data, while testing is made up of the remaining ⅓ data points.

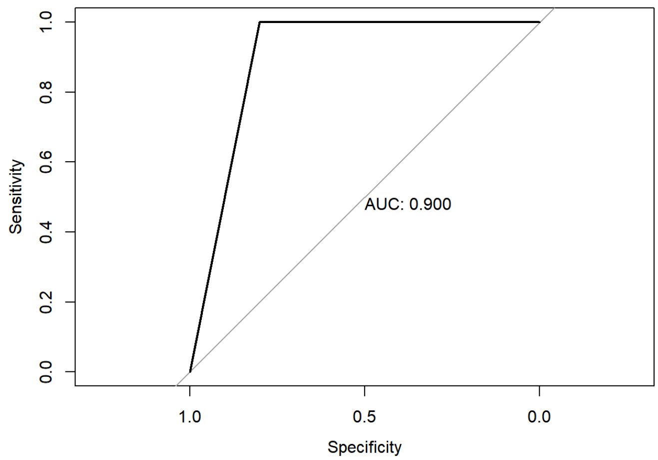
The first model to be tested against accuracy prediction was logistic regression. It only incorrectly assigned 9 patients, which gave an accuracy of 92.97% and an AUC of 0.982.

p	0	1
0	59	3
1	6	60



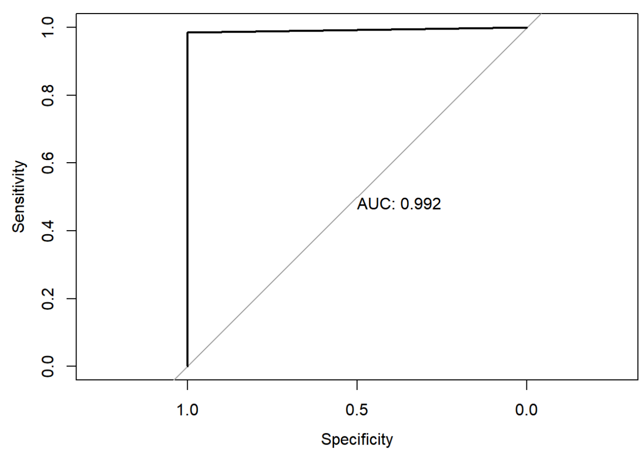
Next was Linear Discriminant Analysis (LDA) which only misassigned 13 patients, giving an accuracy of 89.84% and an AUC of 0.90. It worked well but not as well as logistic regression.

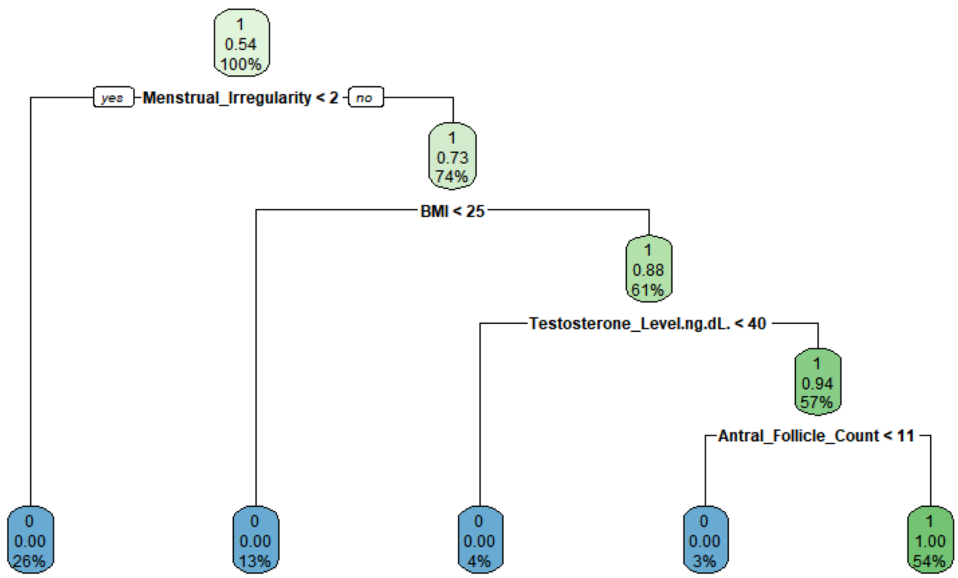
p	0	1
0	52	0
1	13	63



The next prediction model was a decision tree. This was the best performing model as it only misassigned 1 patient out of 382, giving this model a 99.21% accuracy and an AUC of 0.992. I believe this is the best performer since diagnosis PCOS is very much like a flow chart in itself, similar to that of the decision tree.

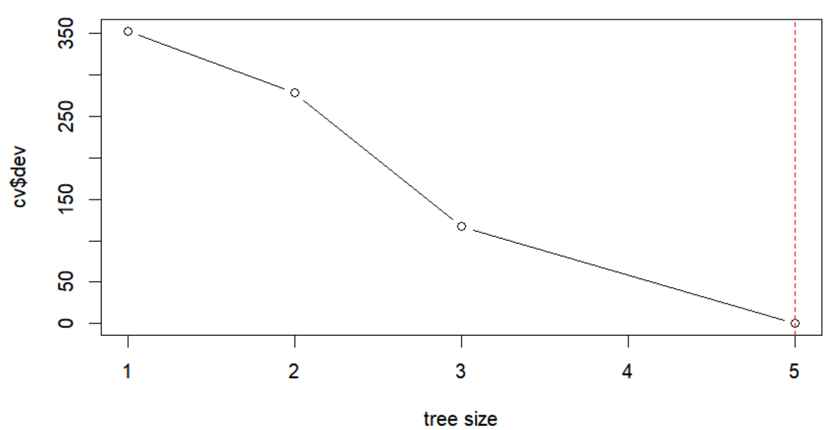
p	0	1
0	65	1
1	0	62





This tree shows that 26% of patients who did have menstrual irregularity but low BMI immediately are not diagnosed with PCOS. Also it says that 54% of the split data that did have a PCOS diagnosis did not have menstrual irregularity but high BMI, testosterone level, and antral follicle count. This is a bit surprising since menstrual irregularity is a symptom of having PCOS.

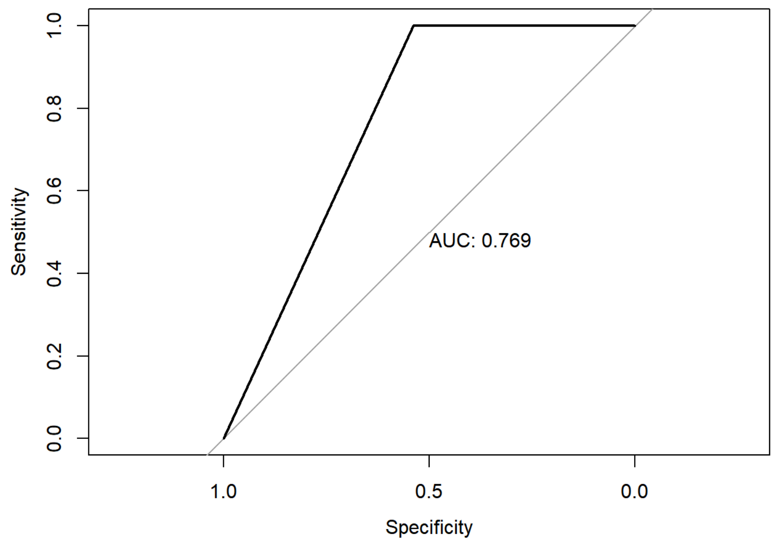
Continuing on with trees, cross-validation was conducted to see the optimal amount of tree leaves, or nodes, for accurate predicting. The optimal amount of leaves that gives the lowest error score was 5 leaves. Since the original tree already has 5 nodes, there was no need to prune the tree any further. This makes sense since there are already a small number of predictors to begin with, and any pruning would lead to a model that is too simple.



Along with decision tree modeling, random forest and bagging were conducted. These two models gave the same exact outcome as the decision tree: misassigning 1 patient, a 99.21% accuracy rate, and an AUC of 99.2. This could be because the model has only 4 predictors to begin with, so there weren't many splits that could possibly take place in random forest and bagging.

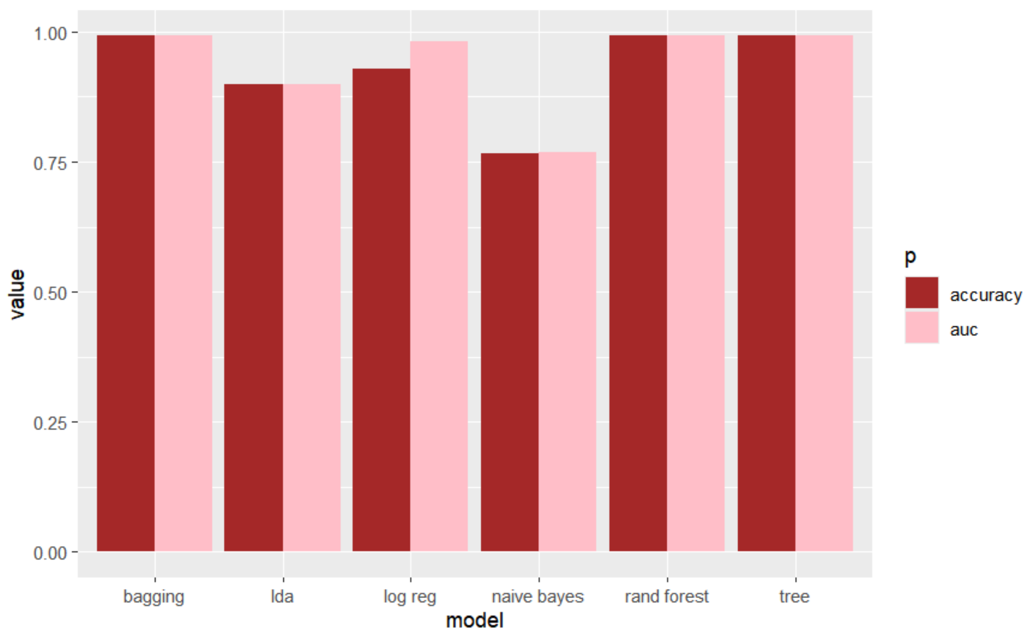
Lastly, Naive Bayes modeling was conducted on the split data and gave an accuracy rate of 76.6% and an AUC of 0.769. This model was by far the worst at predicting as it misassigned 30 students.

p	0	1
0	35	0
1	30	63



Model Comparison and Conclusion

Comparing the models side-by-side will give a better idea of the differences in their performance.



Naive Bayes performed the worst as it had the lowest accuracy and AUC. All of the other models had very high accuracy, with all of them exceeding 80% and going as close to 99%. The best were decision tree, random forest, and bagging, as they all had the same results.

I believe the AUC was so high for all these models because we were using balanced data. If the AUC were to be tested on the raw data where there was about a 2:8 ratio of those with a PCOS diagnosis and those who do not, the AUC would be much lower since it would be harder to tell between the two classes, as there would be biasedness.

Although these models only used 4 predictors, they performed extremely well for such simple conditions, indicating this data set is great for predicting PCOS diagnosis in women!